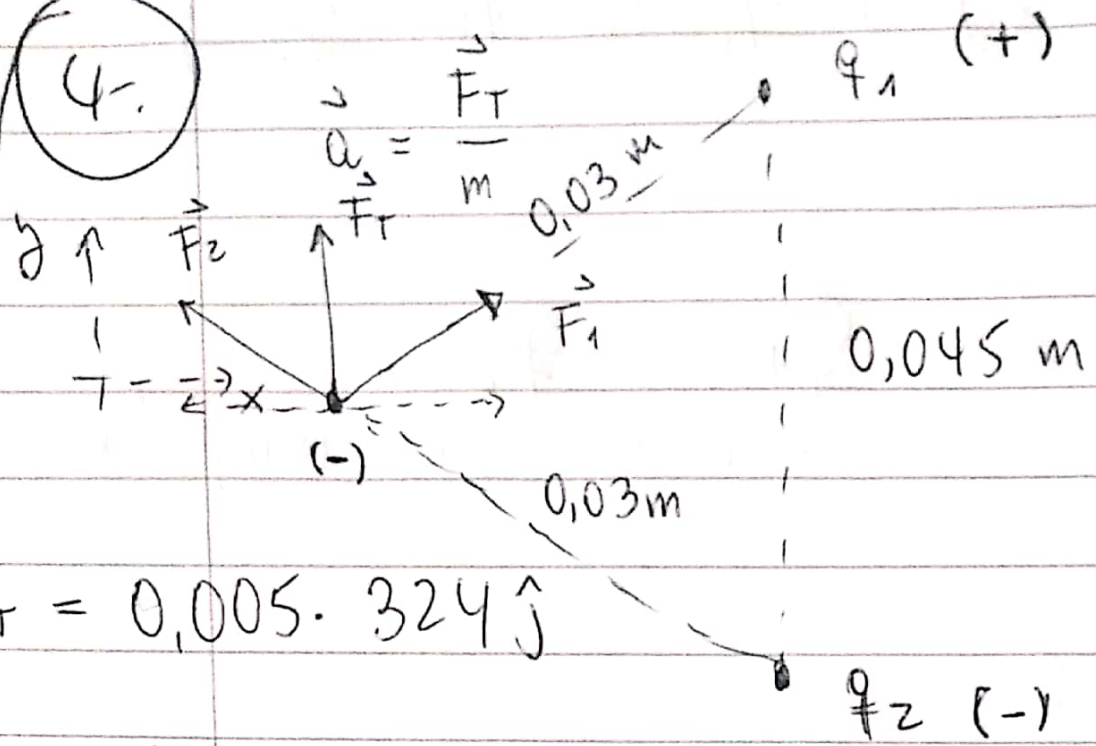
4-

$$\vec{a} = \frac{\vec{F}_T}{m}$$



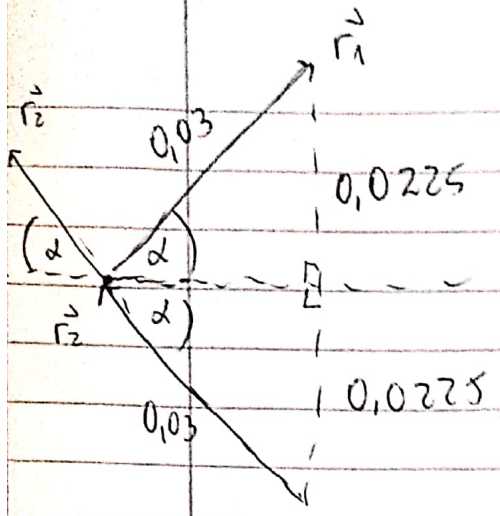
$$\vec{F}_1 = K \frac{Q q_2}{(0{,}03)^3} \vec{r}_1$$

$$\vec{F}_2 = K \frac{Q q_1}{(0{,}03)^3} \vec{r}_2$$

$$\vec{F}_T = 0{,}005 \cdot 324 \hat{j}$$

$$= 1{,}62 \hat{j} \ [N]$$

Escaneado con CamScanner



$$\operatorname{sen}\alpha = \frac{0{,}0225}{0{,}03}$$

$$\alpha = 48{,}59^\circ$$

$$\vec{r}_1 = 0{,}03\cos\alpha\,\hat{\imath} + 0{,}03\operatorname{sen}\alpha\,\hat{\jmath}$$

$$\vec{r}_2 = 0{,}03\cos\alpha\,(-\hat{\imath}) + 0{,}03\operatorname{sen}\alpha\,\hat{\jmath}$$

$$\Rightarrow \vec{F}_1 = 9\cdot10^9\,\frac{(1{,}75\cdot10^{-6})\,q_1}{(0{,}03)^3}\,0{,}03(\cos\alpha\,\hat{\imath} + \operatorname{sen}\alpha\,\hat{\jmath})$$

$$= 1{,}158\cdot10^7\,q_1\,\hat{\imath} + 1{,}312\cdot10^6\,q_1\,\hat{\jmath}$$

$$\vec{F}_2 = 9\cdot10^9\,\frac{(1{,}75\cdot10^{-6})\,q_2}{(0{,}03)^3}\,0{,}03(\cos\alpha\,(-\hat{\imath}) + \operatorname{sen}\alpha\,\hat{\jmath})$$

$$= 1{,}158\cdot10^7\,q_2\,(-\hat{\imath}) + 1{,}312\cdot10^6\,q_2\,\hat{\jmath}$$

$$\vec{F}_1 + \vec{F}_2 = \vec{F}_T$$

$$= 1{,}62\,\hat{\jmath}$$

$\Rightarrow q_1 = q_2$

se cancelan las comp. "x".

$$2\cdot1{,}312\cdot10^6\,q_1 = 1{,}62$$

$$\Rightarrow q_1 = 6{,}174\cdot10^{-7}\,C$$

$$\Rightarrow q_1 = 61{,}74\,\mu C \;//$$

Escaneado con CamScanner